

CCS335 – CLOUD COMPUTING

UNIT I CLOUD ARCHITECTURE MODELS AND INFRASTRUCTURE

Cloud Architecture: System Models for Distributed and Cloud Computing – NIST Cloud Computing Reference Architecture – Cloud deployment models – Cloud service models; Cloud Infrastructure: Architectural Design of Compute and Storage Clouds – Design Challenges.

Cloud Architecture: System Models for Distributed and Cloud Computing

Cloud refers to a Network or Internet. Cloud is something, which is present at remote location. Cloud can provide services over network, that is, on public networks or on private networks, that is, Wide Area Networks (WANs), Local Area Networks (LANs), or Virtual Private Networks (VPNs). Applications such as e-mail, web conferencing, customer relationship management (CRM), all run in cloud.



Figure 1.1 Examples of Cloud Computing

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computer resources (networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.



1. SYSTEM MODELS FOR DISTRIBUTED AND CLOUD COMPUTING

- Distributed and cloud computing systems are
 - Built over a large number of autonomous computer nodes.
 - Interconnected by SANs, LANs, or WANs in a hierarchical manner.
 - A storage area network (SAN) is a dedicated high-speed network or subnetwork that interconnects and presents shared pools of storage devices to multiple servers.
 - LAN switches - connect hundreds of machines as a working cluster.
 - WAN - connect many local clusters to form a very large cluster of clusters.

Classification of Distributed Computing

- Clusters of Cooperative Computers
- Peer-to-Peer Networks
- Grid Computing
- Cloud Computing over the Internet

Clusters of Cooperative Computers

Clustering means that multiple servers are grouped together to achieve the same service.

A computing cluster consists of interconnected stand-alone computers which work cooperatively as a single integrated computing resource.

In the past, clustered computer systems have demonstrated impressive results in handling heavy workloads with large data sets.

Cluster Architecture

The architecture of a typical server cluster built around a low-latency, high bandwidth interconnection network. This network can be as simple as a SAN (e.g., Myrinet) or a LAN (e.g., Ethernet).

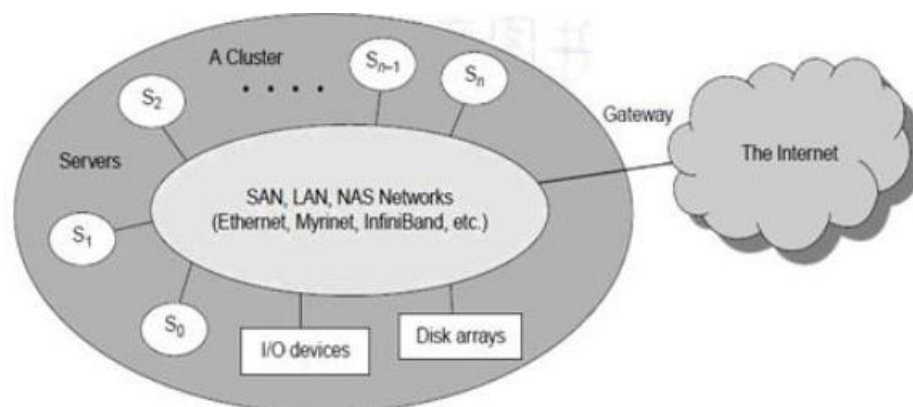


Figure 1 Clusters of Servers

- To build a larger cluster with more nodes, the interconnection network can be built with multiple levels of Gigabit Ethernet, or InfiniBand switches.
- Through hierarchical construction using a SAN, LAN, or WAN, one can build scalable clusters with an increasing number of nodes. The cluster is connected to the Internet via a virtual private network (VPN) gateway.

- The gateway IP address locates the cluster. The system image of a computer is decided by the way the OS manages the shared cluster resources.

Most clusters have loosely coupled node computers. All resources of a server node are managed by their own OS. Thus, most clusters have multiple system images as a result of having many autonomous nodes under different OS control.

Single-System Image (SSI)

- Ideal cluster should merge multiple system images into a single-system image (SSI).
- Cluster designers desire a cluster operating system or some middleware to support SSI at various levels, including the sharing of CPUs, memory, and I/O across all cluster nodes.
- An SSI is an illusion created by software or hardware that presents a collection of resources as one integrated, powerful resource.
- SSI makes the cluster appear like a single machine to the user. A cluster with multiple system images is nothing but a collection of independent computers.

Advantages of Cluster Computing

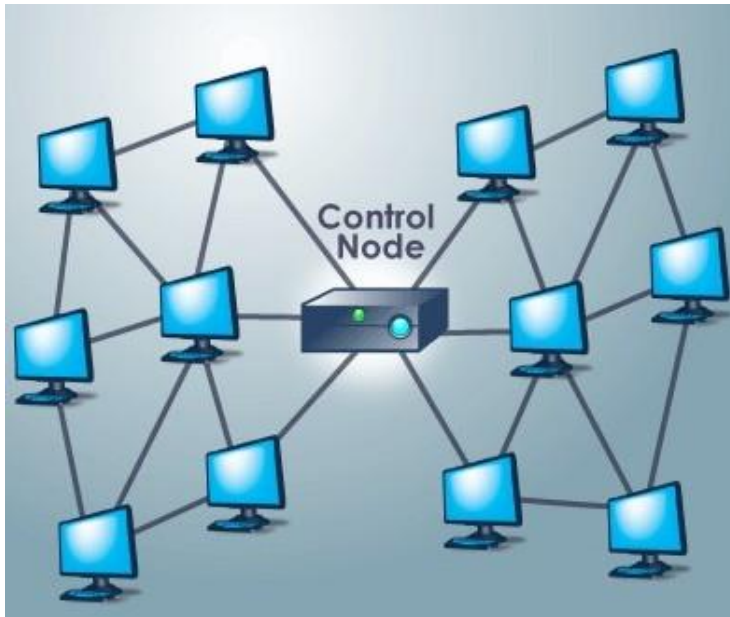
1. High Performance
2. Easy to manage
3. Scalable
4. Expandability
5. Availability
6. Flexibility
7. Cost-effectiveness
8. Distributed applications

Disadvantages of Cluster Computing

1. High cost.
2. The problem is finding the fault.
3. More space is needed.
4. The increased infrastructure is needed.
5. In distributed systems, it is challenging to provide adequate security because both the nodes and the connections must be protected.

Grid Computing

- Grid Computing is a subset of distributed computing.
- In grid computing, the subgroup consists of distributed systems, which are often set up as a network of computer systems, each system can belong to a different administrative domain and can differ greatly in terms of hardware, software, and implementation network technology.



- The different department has different computer with different OS to make the control node present which helps different computer with different OS to communicate with each other and transfer messages to work.

Grid Computing Infrastructures

- An infrastructure that couples computers, software/middleware, special instruments, and people and sensors together.
- Constructed across LAN, WAN, or Internet backbone networks at a regional, national, or global scale.
- Mainly uses workstations, servers, clusters, and supercomputers.
- Personal computers, laptops, and PDAs can be used as access devices to a grid system.
- Industrial grid platform development by IBM, Microsoft, Sun, HP, Dell, Cisco

Advantages of Grid Computing

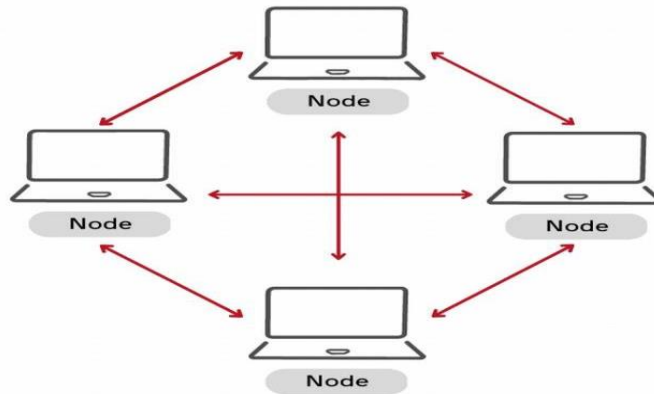
1. Can solve bigger and more complex problems in a shorter time frame. Easier collaboration with other organizations and better use of existing equipment.
2. Existing hardware is used to the fullest.
3. Collaboration with organizations made easier

Disadvantages of Grid Computing

1. You may need a fast connection between computer resources.
2. Licensing on many servers can be prohibitive for some applications.

Peer To Peer Network (P2P)

- Peer to Peer network is group of computers each of which acts as a node for sharing files within the group.
- It allows people to share the files with each other without the need of centralized server.



- The P2P architecture offers a distributed model of networked systems.
- A P2P network is client-oriented instead of server-oriented.
- P2P systems are introduced at the
 - physical network
 - overlay networks at the logical levels

Physical Network:

- The participating peers form the physical network at any time.
- Unlike the cluster or grid, a P2P network does not use a dedicated interconnection network.
- The physical network is simply an ad hoc network formed at various Internet domains randomly using the TCP/IP and NAI protocols

Overlay Network

- Based on communication or file-sharing needs, the peer IDs form an overlay network at the logical level.
- This overlay is a virtual network formed by mapping each physical machine with its ID, logically, through a virtual mapping.
- When a new peer joins the system, its peer ID is added as a node in the overlay network and is removed from the overlay network automatically when it leaves.
- Therefore, it is the P2P overlay network that characterizes the logical connectivity among the peers.
- Two types of overlay networks:
 - unstructured and structured

Cloud Computing over the Internet

- A cloud is a pool of virtualized computer resources.
- A cloud can host a variety of different workloads, including batch-style backend jobs and interactive and user-facing applications.

- A cloud allows workloads to be deployed and scaled out quickly through rapid provisioning of virtual or physical machines.
- The cloud supports redundant, self-recovering, highly scalable programming models that allow workloads to recover from many unavoidable hardware/software failures.
- Finally, the cloud system should be able to monitor resource use in real time to enable rebalancing of allocations when needed.

a. Internet Clouds

- Cloud computing applies a virtualized platform with elastic resources on demand by provisioning hardware, software, and data sets dynamically. The idea is to move desktop computing to a service-oriented platform using server clusters and huge databases at data centres.
- Cloud computing leverages its low cost and simplicity to benefit both users and providers.
- Machine virtualization has enabled such cost-effectiveness. Cloud computing intends to satisfy many user applications simultaneously.

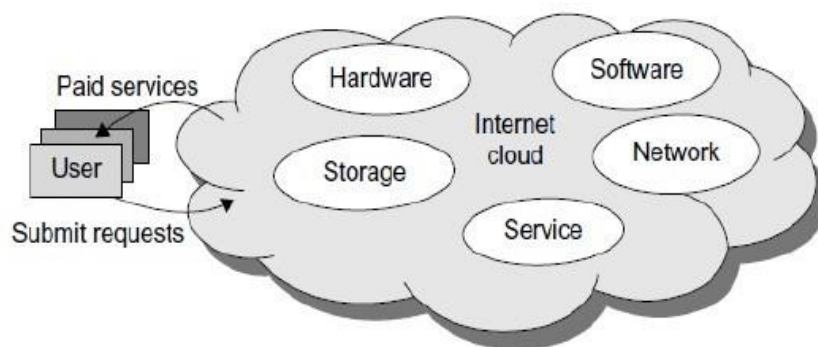


Figure: Internet Cloud

b. The Cloud Landscape

- The cloud ecosystem must be designed to be secure, trustworthy, and dependable.
- Some computer users think of the cloud as a centralized resource pool.
- Others consider the cloud to be a server cluster which practices distributed computing over all the servers.
- Traditionally, a distributed computing system tends to be owned and operated by an autonomous administrative domain (e.g., a research laboratory or company) for on-premises computing needs.
- Cloud computing as an on-demand computing paradigm resolves or relieves us from these problems.

2. NIST Cloud Computing Reference Architecture

Definition

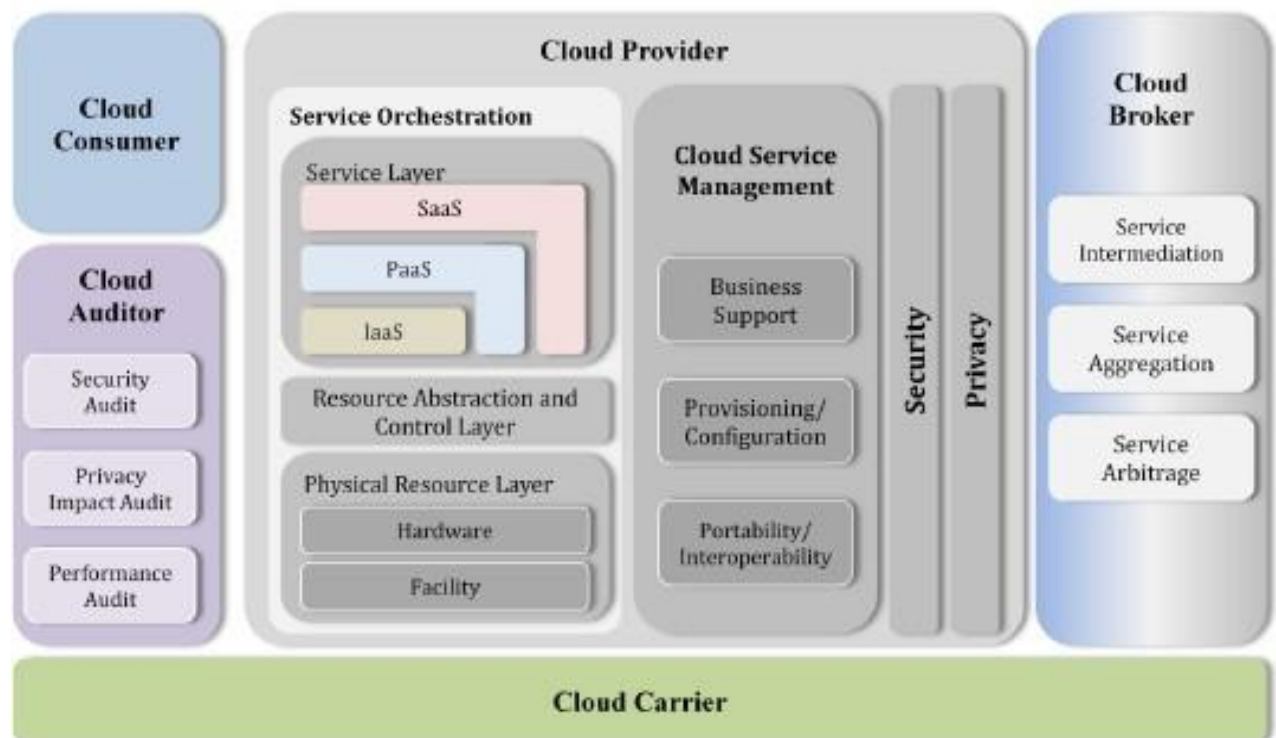
"The National Institute of Standards and Technology (NIST) defines cloud computing as a "pay-per-use model for enabling available, convenient and on- demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

The NIST cloud computing reference architecture defines five major actors:

- cloud consumer
- cloud provider
- cloud carrier
- cloud auditor
- cloud broker

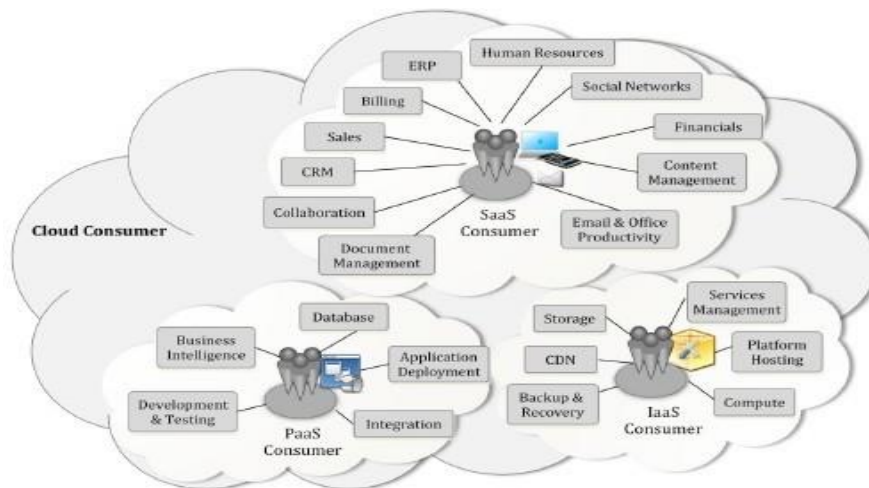
Each actor is an entity (a person or an organization) that participates in a transaction or process and/or performs tasks in cloud computing.

NIST Cloud Reference Model



Cloud Consumer

- The cloud consumer is the principal stakeholder for the cloud computing service.
- A cloud consumer represents a person or organization that maintains a business relationship with, and uses the service from a cloud provider.
- Cloud consumers need SLAs to specify the technical performance requirements fulfilled by a cloud provider.
- SLAs can cover terms regarding the quality of service, security, remedies for performance failures.



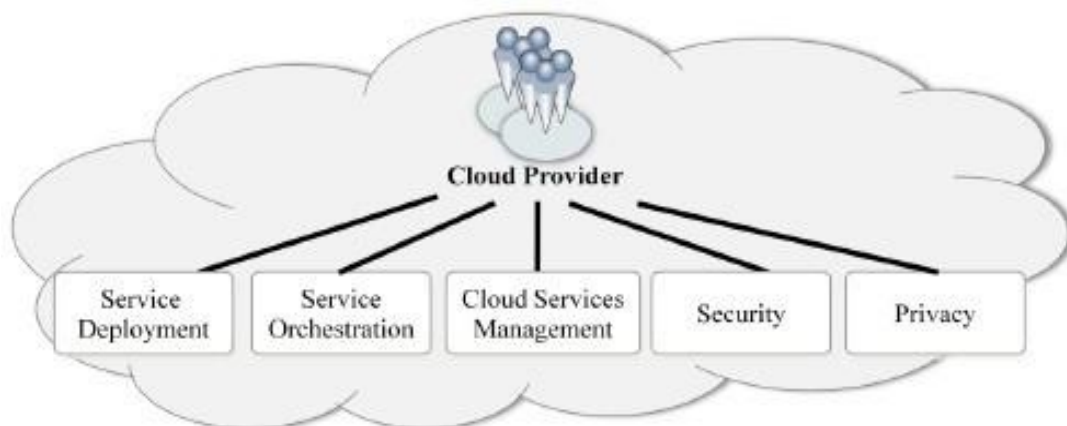
Example Services Available to a Cloud Consumer

Cloud Services on Cloud Consumer

- The consumers of SaaS can be organizations that provide their members with access to software applications, end users who directly use software applications, or software application administrators who configure applications for end users.
- Cloud consumers of PaaS can employ the tools and execution resources provided by cloud providers to develop, test, deploy and manage the applications hosted in a cloud environment.
- Consumers of IaaS have access to virtual computers, network-accessible storage, network infrastructure components, and other fundamental computing resources on which they can deploy and run arbitrary software.

Cloud Provider

- A cloud provider is a person, an organization; it is the entity responsible for making a service available to interested parties.
- A Cloud Provider acquires and manages the computing infrastructure required for providing the services, runs the cloud software that provides the services, and makes arrangement to deliver the cloud services to the Cloud Consumers through network access.



5 Major Activities of Cloud Provider

Service on Cloud Provider

- For Software as a Service, the cloud provider deploys, configures, maintains and updates the operation of the software applications on a cloud infrastructure so that the services are provisioned at the expected service levels to cloud consumers.
- The provider of SaaS assumes most of the responsibilities in managing and controlling the applications and the infrastructure, while the cloud consumers have limited administrative control of the applications.
- For PaaS, the Cloud Provider manages the computing infrastructure for the platform and runs the cloud software that provides the components of the platform, such as runtime software execution stack, databases, and other middleware components.
- For IaaS, the Cloud Provider acquires the physical computing resources underlying the service, including the servers, networks, storage and hosting infrastructure

Cloud Auditor

- A cloud auditor is a party that can perform an independent examination of cloud service controls with the intent to express an opinion thereon.
- A cloud auditor can evaluate the services provided by a cloud provider in terms of security controls, privacy impact, performance, etc.

Cloud Broker

- As cloud computing evolves, the integration of cloud services can be too complex for cloud consumers to manage.
- A cloud consumer may request cloud services from a cloud broker, instead of contacting a cloud provider directly.
- A cloud broker is an entity that manages the use, performance and delivery of cloud services and negotiates relationships between cloud providers and cloud consumers.

Categories of Cloud Broker

- A cloud broker can provide services in three categories
 - **Service Intermediation:** A cloud broker enhances a given service by improving some specific capability and providing value-added services to cloud consumers. The improvement can be managing access to cloud services, identity management, performance reporting, enhanced security, etc.
 - **Service Aggregation:** A cloud broker combines and integrates multiple services into one or more new services. The broker provides data integration and ensures the secure data movement between the cloud consumer and multiple cloud providers.
 - **Service Arbitrage:** Service arbitrage is similar to service aggregation except that the services being aggregated are not fixed. Service arbitrage means a broker has the flexibility to choose services from multiple agencies. The cloud broker, for example, can use a credit-scoring service to measure and select an agency with the

best score.

Cloud Carrier

- A cloud carrier acts as an intermediary that provides connectivity and transport of cloud services between cloud consumers and cloud providers. Cloud carriers provide access to consumers through network, telecommunication and other access devices.
- For example, cloud consumers can obtain cloud services through network access devices, such as computers, laptops, mobile phones, mobile Internet devices (MIDs), etc.

Cloud Computing Architecture

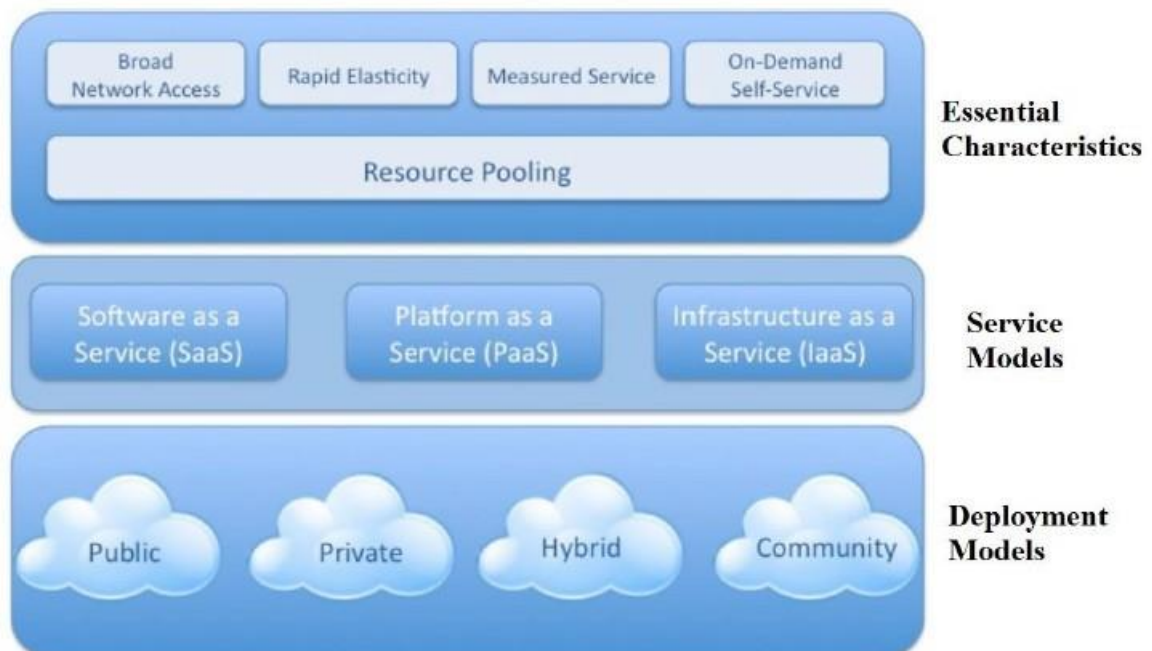
Architecture consists of 3 tiers

- Cloud Deployment Model
- Cloud Service Model
- Essential Characteristics of Cloud Computing

Essential Characteristics of Cloud Computing

1. On-demand self-service.

- A consumer can unilaterally provision computing capabilities such as server time and network storage as needed automatically, without requiring human interaction with a service provider.



1. Figure 1.5 Cloud Computing Architecture

2. Broad network access.

- Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, laptops, and PDAs) as well as other traditional or cloud-based software services.

3. Resource pooling.

- The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.

4. Rapid elasticity.

- Capabilities can be rapidly and elastically provisioned - in some cases automatically - to quickly scale out; and rapidly released to quickly scale in.
- To the consumer, the capabilities available for provisioning often appear to be unlimited and can be purchased in any quantity at any time.

5. Measured service.

- Cloud systems automatically control and optimize resource usage by leveraging a metering capability at some level of abstraction appropriate to the type of service.
- Resource usage can be monitored, controlled, and reported - providing transparency for both the provider and consumer of the service.

3. Cloud Deployment Model

It works as your virtual computing environment with a choice of deployment model depending on how much data you want to store and who has access to the infrastructure.

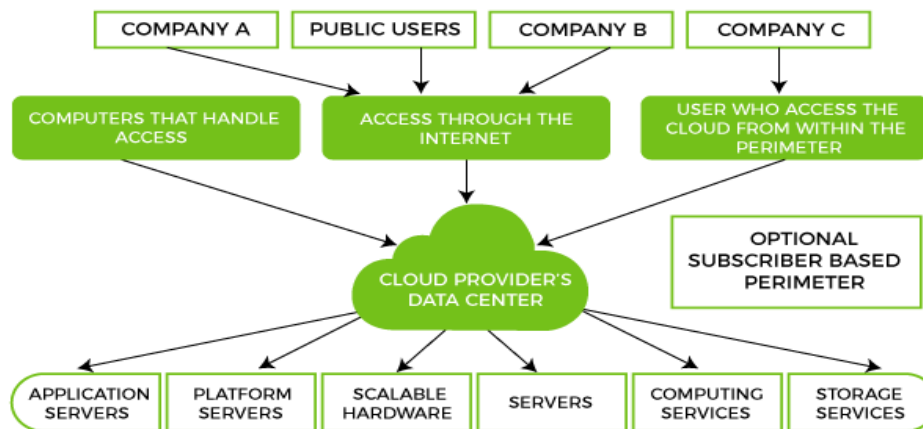
Types of Deployment Model

- Public Cloud
- Private Cloud
- Hybrid Cloud

Public Cloud

- The Public Cloud allows systems and services to be easily accessible to the general public. Public cloud may be less secure because of its openness, e.g., e-mail, Microsoft Azure, AWS, Google Cloud etc.
- Public deployment models in the cloud are perfect for organizations with growing and fluctuating demands.
- It also makes a great choice for companies with low-security concerns. Thus, you pay a cloud service provider for networking services, compute virtualization & storage available on the public internet.
- It is also a great delivery model for the teams with development and testing.
- Its configuration and deployment are quick and easy, making it an ideal choice for test environments.

Public Cloud



Benefits of Public Cloud

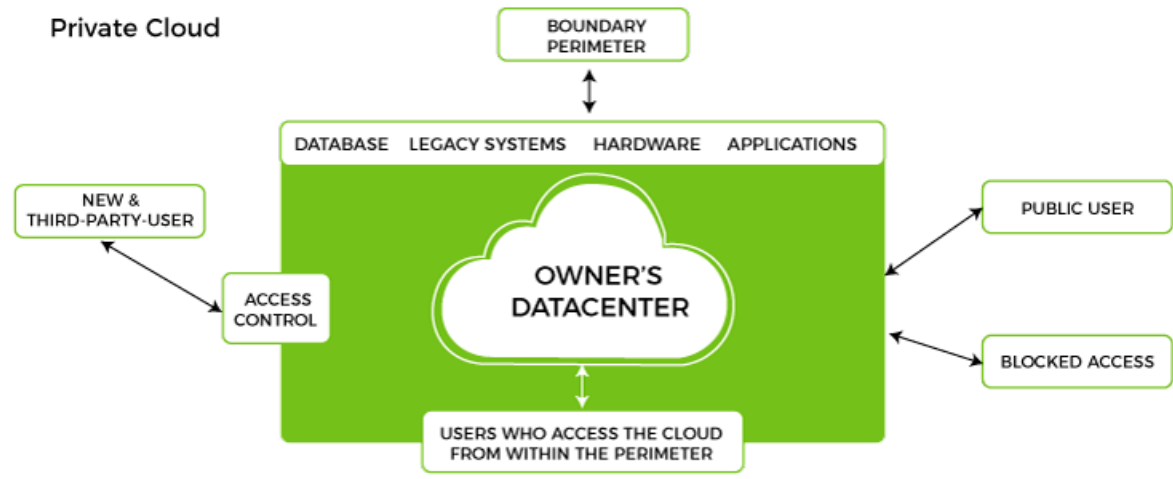
- **Minimal Investment** - As a pay-per-use service, there is no large upfront cost and is ideal for businesses who need quick access to resources
- **No Hardware Setup** - The cloud service providers fully fund the entire infrastructure
- **No Infrastructure Management** - This does not require an in-house team to utilize the public cloud.

Limitations of Public Cloud

- **Data Security and Privacy Concerns** - Since it is accessible to all, it does not fully protect against cyber-attacks and could lead to vulnerabilities.
- **Reliability Issues** - Since the same server network is open to a wide range of users, it can lead to malfunction and outages
- **Service/License Limitation** - While there are many resources you can exchange with tenants, there is a usage cap.

Private Cloud

- The Private Cloud allows systems and services to be accessible within an organization. It offers increased security because of its private nature.
- Companies that look for cost efficiency and greater control over data & resources will find the private cloud a more suitable choice.
- It means that it will be integrated with your data center and managed by your IT team.
- The private cloud offers bigger opportunities that help meet specific organizations' requirements when it comes to customization.



Benefits of Private Cloud

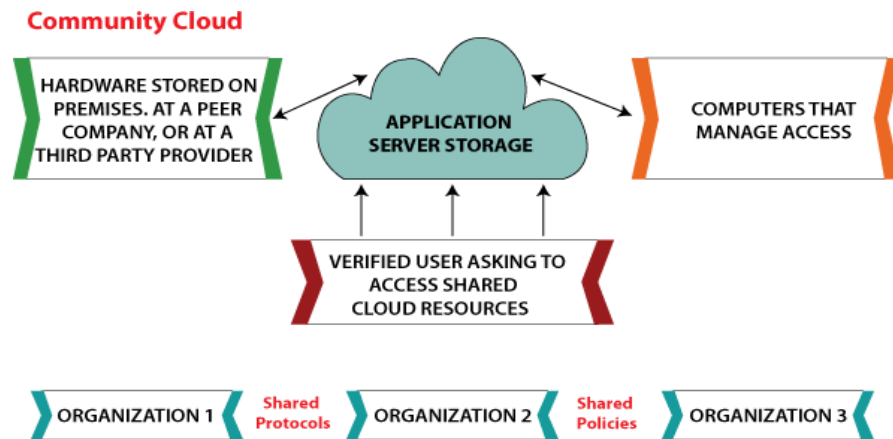
- **Data Privacy** - It is ideal for storing corporate data where only authorized personnel gets access
- **Security** - Segmentation of resources within the same Infrastructure can help with better access and higher levels of security.
- **Supports Legacy Systems** - This model supports legacy systems that cannot access the public cloud.

Limitations of Private Cloud

- **Higher Cost** - With the benefits you get, the investment will also be larger than the public cloud. Here, you will pay for software, hardware, and resources for staff and training.
- **Fixed Scalability** - The hardware you choose will accordingly help you scale in a certain direction
- **High Maintenance** - Since it is managed in-house, the maintenance costs also increase.

Community Cloud

- The Community Cloud allows systems and services to be accessible by group of organizations.
- The community cloud operates in a way that is similar to the public cloud. There's just one difference - it allows access to only a specific set of users who share common objectives and use cases.
- This type of deployment model of cloud computing is managed and hosted internally or by a third-party vendor. However, you can also choose a combination of all three.



Benefits of Community Cloud

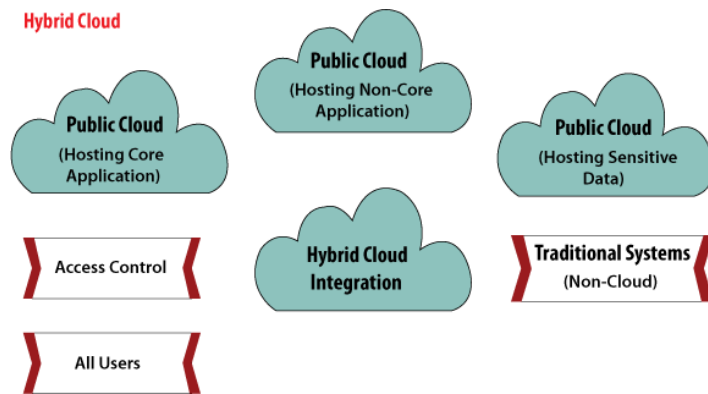
- Smaller Investment - A community cloud is much cheaper than the private & public cloud and provides great performance
- Setup Benefits - The protocols and configuration of a community cloud must align with industry standards, allowing customers to work much more efficiently.

Limitations of Community Cloud

- Shared Resources - Due to restricted bandwidth and storage capacity, community resources often pose challenges.
- Not as Popular - Since this is a recently introduced model, it is not that popular or available across industries

Hybrid Cloud

- The Hybrid Cloud is mixture of public and private cloud.
- However, the critical activities are performed using private cloud while thenon- critical activities are performed using public cloud.
- A hybrid cloud is a combination of two or more cloud architectures.
- While each model in the hybrid cloud functions differently, it is all part ofthe same architecture.
- Further, as part of this deployment of the cloud computing model, the internal or external providers can offer resources.
- Consider a company with critical data will prefer storing on a private cloud, while less sensitive data can be stored on a public cloud. The hybrid cloud is also frequently used for 'cloud bursting'. It means, supposes an organization runs an application on-premises, but due to heavy load, it can burst into the public cloud.



Benefits of Hybrid Cloud

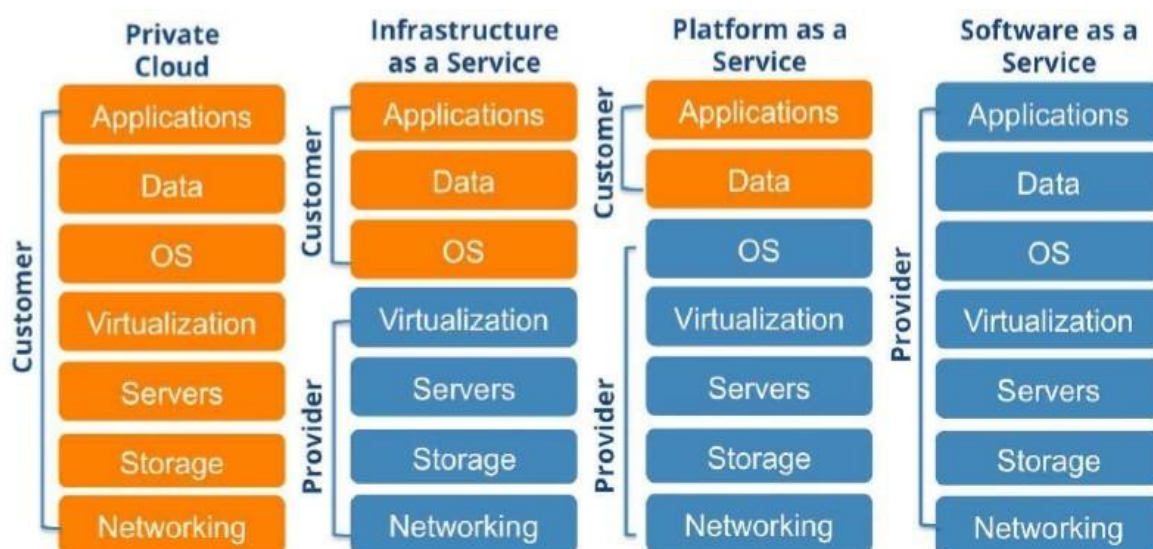
- Cost-Effectiveness - The overall cost of a hybrid solution decreases since it majorly uses the public cloud to store data.
- Security - Since data is properly segmented, the chances of data theft from attackers are significantly reduced.
- Flexibility - With higher levels of flexibility, businesses can create custom solutions that fit their exact requirements

Limitations of Hybrid Cloud

- Complexity - It is complex setting up a hybrid cloud since it needs to integrate two or more cloud architectures
- Specific Use Case - This model makes more sense for organizations that have multiple use cases or need to separate critical and sensitive data

4. Cloud Service Models

- Cloud Software as a Service (SaaS)
- Cloud Platform as a Service (PaaS)
- Cloud Infrastructure as a Service (IaaS)



Cloud Service Model Architecture

Software as a Service (SaaS)

- SaaS is a licensed software offering on the cloud and pay per use
- SaaS is a software delivery methodology that provides licensed multi-tenant access to software and its functions remotely as a Web-based service.
- Customers do not invest on software application programs
- The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure.
- The applications are accessible from various client devices through a thin client interface such as a web browser (e.g., web-based email).
- The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, data or even individual application capabilities, with the possible exception of limited user specific application configuration settings.

Characteristics of SaaS

- Managed from a central location
- Hosted on a remote server
- Accessible over the internet
- Users are not responsible for hardware and software updates. Updates are applied automatically.
- The services are purchased on the pay-as-per-use basis

SaaS providers

- Google Apps, Gmail, Docs, Talk etc
- Microsoft's Hotmail, Sharepoint
- Salesforce
- Yahoo, Facebook

Platform as a Service (PaaS)

- PaaS provides all of the facilities required to support the complete life cycle of building, delivering and deploying web applications and services entirely from the Internet.
- Typically, applications must be developed with a particular platform in mind
 - Multi-tenant environments
- The capability provided to the consumer is to deploy onto the cloud infrastructure consumer created or acquired applications created using programming languages and tools supported by the provider.
- The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly application hosting environment configurations.

Characteristics of PaaS

- Accessible to various users via the same development application.
- Integrates with web services and databases.
- Builds on virtualization technology, so resources can easily be scaled up or down as per the organization's need.
- Support multiple languages and frameworks.
- Provides an ability to Auto-scale.

PaaS providers

- Google App Engine - Python, Java, Eclipse
- Microsoft Azure - .Net, Visual Studio
- Sales Force - Apex, Web wizard
- TIBCO,
- VMware,
- Zoho

Infrastructure as a Service (IaaS)

- IaaS is the delivery of technology infrastructure (mostly hardware) as an on demand, scalable service
 - Usually billed based on usage
 - Usually, multi-tenant virtualized environment
 - Can be coupled with Managed Services for OS and application support
 - User can choose his OS, storage, deployed app, networking components
- The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources.
- Consumer is able to deploy and run arbitrary software, which may include operating systems and applications.
- The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of select networking components (e.g., host firewalls).

Characteristics of IaaS

- Resources are available as a service
- Services are highly scalable
- Dynamic and flexible
- GUI and API-based access
- Automated administrative tasks

IaaS providers

- Amazon Elastic Compute Cloud (EC2)- Each instance provides 1-20 processors, up to 16 GB RAM, 1.69 TB storage
- RackSpace Hosting- Each instance provides 4 core CPU, up to 8 GB RAM, 480 GB storage

- Joyent Cloud- Each instance provides 8 CPUs, upto 32 GB RAM, 48 GB storage
- Go Grid- Each instance provides 1-6 processors, upto 15 GB RAM, 1.69TB storage

5. Cloud Infrastructure: Architectural Design of Compute and Storage Clouds

Generic Cloud Architecture Design

An Internet cloud is envisioned as a public cluster of servers provisioned on demand to perform collective web services or distributed applications using data-center resources.

- Cloud Platform Design Goals
- Enabling Technologies for Clouds
- A Generic Cloud Architecture

Cloud Platform Design Goals

- Scalability
- Virtualization
- Efficiency
- Reliability
- Security

Cloud management receives the user request and finds the correct resources. Cloud calls the provisioning services which invoke the resources in the cloud. Cloud management software needs to support both physical and virtual machines.

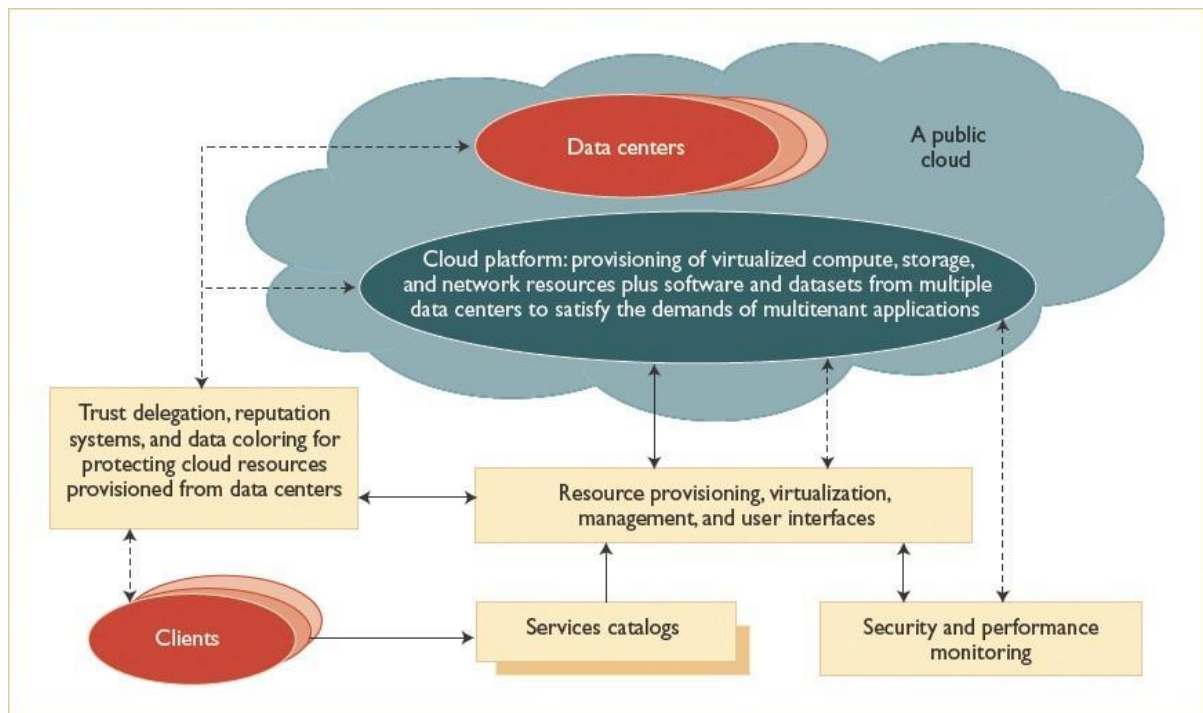
Enabling Technologies for Clouds

- Cloud users are able to demand more capacity at peak demand, reduce costs, experiment with new services, and remove unneeded capacity.
- Service providers can increase system utilization via multiplexing, virtualization and dynamic resource provisioning.
- Clouds are enabled by the progress in hardware, software and networking technologies.
- In the hardware area, the rapid progress in multicore CPUs, memory chips, and disk arrays has made it possible to build faster data centers with huge amounts of storage space.
- Resource virtualization enables rapid cloud deployment and disaster recovery.
- Service-oriented architecture (SOA) also plays a vital role.

A Generic Cloud Architecture

- The Internet cloud is envisioned as a massive cluster of servers.
- Servers are provisioned on demand to perform collective web services using data-center resources.
- The cloud platform is formed dynamically by provisioning or deprovisioning servers, software, and database resources.
- Servers in the cloud can be physical machines or VMs.

- User interfaces are applied to request services.

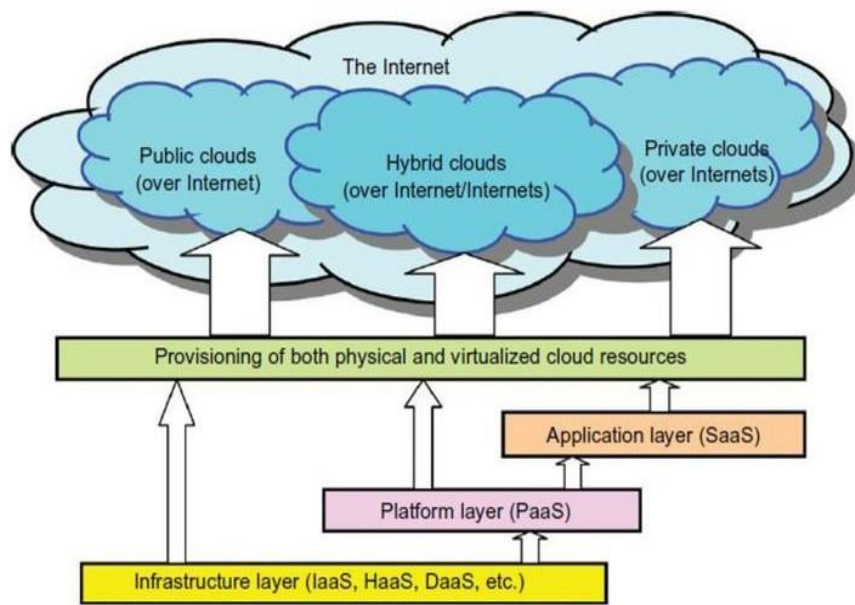


- The cloud computing resources are built into the data centers.
- Data centers are typically owned and operated by a third-party provider. Consumers do not need to know the underlying technologies
- In a cloud, software becomes a service.
- Cloud demands a high degree of trust of massive amounts of data retrieved from large data centers.
- The software infrastructure of a cloud platform must handle all resource management and maintenance automatically.
- Software must detect the status of each node server joining and leaving.
- Cloud computing providers such as Google and Microsoft, have built a large number of data centers.
- Each data center may have thousands of servers.
- The location of the data center is chosen to reduce power and cooling costs.

Layered Cloud Architectural Development

The architecture of a cloud is developed at three layers

- Infrastructure Layer
- Platform Layer
- Application Layer



- Implemented with virtualization and standardization of hardware and software resources provisioned in the cloud.
- The services to public, private and hybrid clouds are conveyed to users through networking support

Infrastructure Layer

- Foundation for building the platform layer.
- Built with virtualized compute, storage, and network resources.
- Provide the flexibility demanded by users.
- Virtualization realizes automated provisioning of resources and optimizes the infrastructure management process.

Platform Layer

- Foundation for implementing the application layer for SaaS applications.
- Used for general-purpose and repeated usage of the collection of software resources.
- Provides users with an environment to develop their applications, to test operation flows, and to monitor execution results and performance.
- The platform should be able to assure users that they have scalability, dependability, and security protection

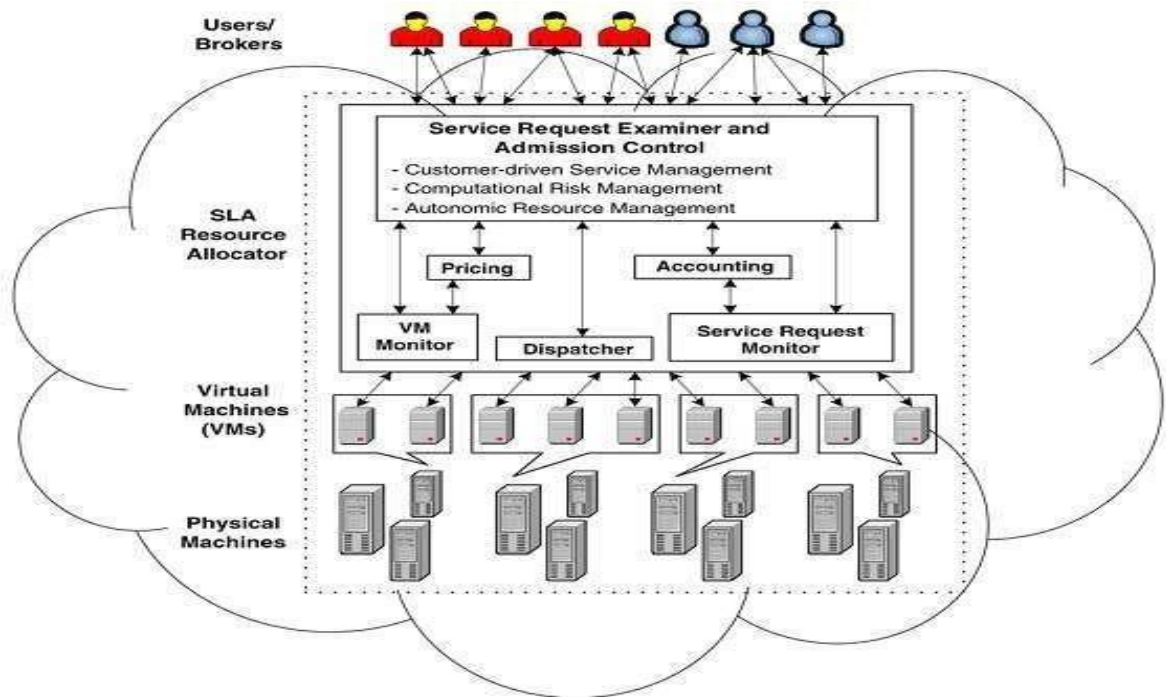
Application Layer

- Collection of all needed software modules for SaaS applications.
- Service applications in this layer include daily office management work, such as information retrieval, document processing, and authentication services.
- The application layer is also heavily used by enterprises in business marketing and sales, consumer relationship management (CRM) and financial transactions.
- Not all cloud services are restricted to a single layer.

- Many applications may apply resources at mixed layers.
- Three layers are built from the bottom up with a dependence relationship.

Market-Oriented Cloud Architecture

- High-level architecture for supporting market-oriented resource allocation in a cloud computing environment.
- Users or brokers acting on user's behalf submit service requests to the data center.



- The SLA resource allocator acts as the interface between the data center/cloud service provider and external users.
- When a service request is first submitted, the service request examiner interprets the submitted request for QoS requirements.
- Accept or Reject the request.
- **VM Monitor:** Latest status information regarding resource availability.
- **Service Request Monitor:** Latest status information workload processing
- **Pricing mechanism:** Decides how service requests are charged.
- **Accounting mechanism:** Maintains the actual usage of resources by requests to compute the final cost.
- VM Monitor mechanism keeps track of the availability of VMs and their resource entitlements.
- Dispatcher starts the execution of accepted service requests on allocated VMs.
- Service Request Monitor mechanism keeps track of the execution progress of service requests.
- Multiple VMs can be started and stopped on demand.

Quality of Service Factors QoS parameters

- Time
- Cost
- Reliability
- Trust/security

QoS requirements cannot be static and may change over time.

6. Design Challenges

Six open challenges in cloud architecture development

- Service Availability and Data Lock-in Problem.
- Data Privacy and Security.
- Unpredictable Performance and Bottlenecks.
- Distributed Storage and Widespread Bugs
- Cloud Scalability, Interoperability and Standardization.
- Software Licensing

Challenge 1: Service Availability and Data Lock-in Problem**Availability**

Service Availability in Cloud might be affected because of

- Single Point Failure Distributed
- Denial of Service
- Single Point Failure
 - Depending on single service provider might result in failure.
 - In case of single service providers, even if company has multiple data centers located in different geographic regions, it may have **common software infrastructure and accounting systems**.

Solution:

- Multiple cloud providers may provide more protection from failures and they provide High Availability (HA)
- Multiple cloud Providers will rescue the loss of all data.

Distributed Denial of service (DDoS) attacks.

- Cyber criminals, attack target websites and online services and make services unavailable to users.
- DDoS tries to overwhelm (disturb) the services unavailable to user by having more traffic than the server or network can accommodate.

Solution:

- Some SaaS providers provide the opportunity to defend against DDoS attacks by using quick scale-ups.

Customers cannot easily extract their data and programs from one site to run on another.

Solution:

- Have standardization among service providers so that customers can deploy(install) services and data across multiple cloud providers.

Data Lock-in

- It is a situation in which a customer using service of a provider cannot be moved to another service provider because technologies used by a provider will be incompatible with other providers?
- This makes a customer dependent on a vendor for services and makes customer unable to use service of another vendor.

Solution:

- Have standardization (in technologies) among service providers so thatcustomers can easily move from a service provider to another.

Challenge 2: Data Privacy and Security Concerns

Cloud services are prone to attacks because they are accessed through internet.Security is given by o Storing the encrypted data in to cloud.

- Firewalls, filters. Cloud environment attacks include
- Guest hopping
- Hijacking
- VM rootkits.

Guest Hopping: Virtual machine hyper jumping (VM jumping) is an attack method that exploits (make use of) hypervisor's weakness that allows a virtual machine (VM) to be accessed from another. Hijacking: Hijacking is a type of network security attack in which the attacker takes control of a communication.

VM Rootkit: is a collection of malicious (harmful) computer software, designed to enable access to a computer that is not otherwise allowed.

A **man-in-the-middle (MITM)** attack is a form of eavesdropping(Spy) where communication between two users is monitored and modified by an unauthorized party.

o Man-in-the-middle attack may take place **during VM migrations** [virtual machine (VM) migration - VM is moved from one physical host to another host].

Passive attacks steal sensitive data or passwords.

Active attacks may manipulate (control) kernel data structures which will causemajor damage to cloud servers.

Challenge 3: Unpredictable Performance and Bottlenecks

- Multiple VMs can share CPUs and main memory in cloud computing, butI/O sharing is problematic.
- Internet applications continue to become more data-intensive (handles hugeamount of data).
- Handling huge amount of data (data intensive) is a bottleneck in cloudenvironment.

- Weak Servers that does not provide data transfers properly must be removed from cloud environment

Challenge 4: Distributed Storage and Widespread Software Bugs

- The database is always growing in cloud applications.
- There is a need to create a storage system that meets this growth.
- This demands the design of efficient distributed SANs (Storage Area Network of Storage devices).
- Data centres must meet
 - Scalability
 - Data durability
 - HA(High Availability)
 - Data consistence
- Bug refers to errors in software.
- Debugging must be done in data centres.

Challenge 5: Cloud Scalability, Interoperability and Standardization Cloud Scalability

- Cloud resources are scalable.
- Cost increases when storage and network bandwidth scaled(increased).

Interoperability

- Open Virtualization Format (OVF) describes an open, secure, portable, efficient, and extensible format for the packaging and distribution of VMs.
- OVF defines a transport mechanism for VM, that can be applied to different virtualization platforms.

Standardization

- Cloud standardization, should have ability for virtual machine to run on any virtual platform.

Challenge 6: Software Licensing and Reputation Sharing

- Cloud providers can use both pay-for-use and bulk-use licensing schemes to widen the business coverage.
- Cloud providers must create reputation-guarding services similar to the “trusted e-mail” services.
- Cloud providers want legal liability to remain with the customer, and vice versa.

***** All the Best *****